

Chapter

5

Normal Probability Distributions

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

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Chapter Outline

- 5.1 Introduction to Normal Distributions and the Standard Normal Distribution
- 5.2 Normal Distributions: Finding Probabilities
- 5.3 Normal Distributions: Finding Values
- 5.4 Sampling Distributions and the Central Limit Theorem
- 5.5 Normal Approximations to Binomial Distributions

Section 5.2

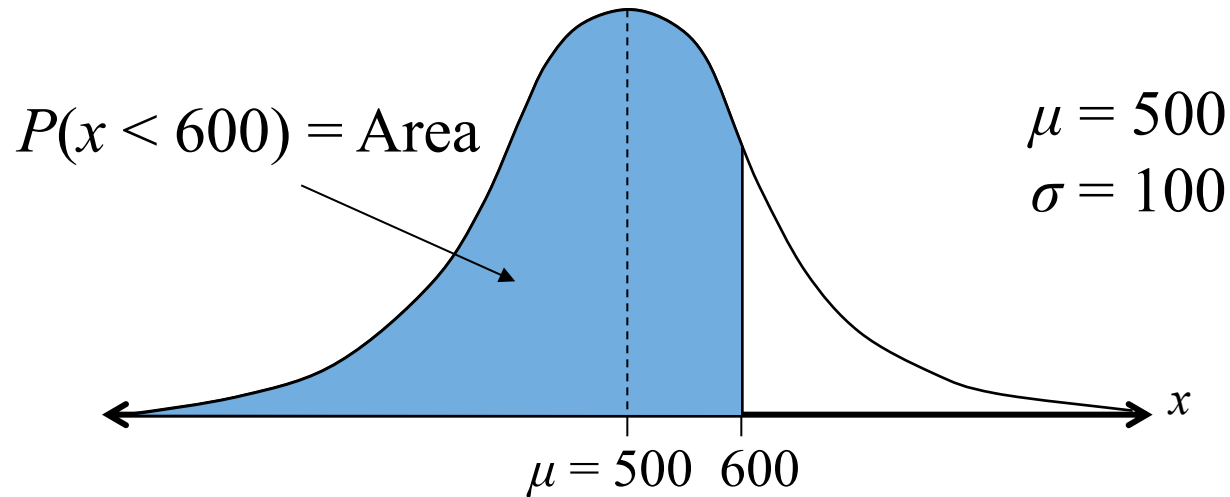
Normal Distributions: Finding Probabilities

Section 5.1 Objectives

- How to find probabilities for normally distributed variables using a table and using technology

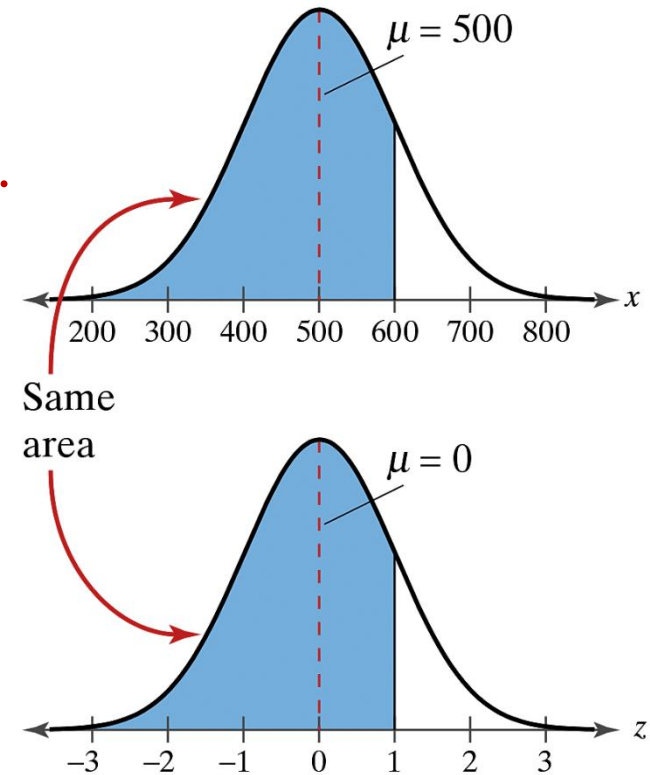
Probability and Normal Distributions

- If a random variable x is normally distributed, you can find the probability that x will fall in a given interval by calculating the area under the normal curve for that interval.



Probability and Normal Distributions

- Consider a normal curve with $\mu = 500$ and $\sigma = 100$ (upper right).
- The value of x one standard deviation above the mean is $\mu + \sigma = 500 + 100 = 600$.
- Now consider the standard normal curve (lower right).
- The value of z one standard deviation above the mean is $\mu + \sigma = 0 + 1 = 1$.
- The z -score of 1 corresponds to an x -value of 600, and areas are not changed with a transformation to a standard normal curve, the shaded areas at the right are equal.



Example: Finding Probabilities for Normal Distributions

A national study found that college students with jobs worked an average of 25 hours per week. The standard deviation is 11 hours. A college student with a job is selected at random. Find the probability that the student works for less than 5 hours per week. Assume that the lengths of time college students work are normally distributed and are represented by the variable x .

(Adapted from Statista)

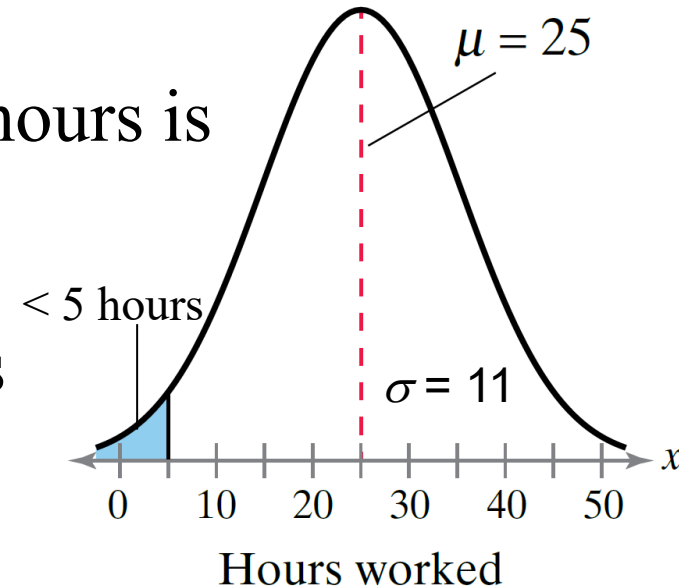
Solution: Finding Probabilities for Normal Distributions

Solution

- The z-score that corresponds to 5 hours is

$$z = \frac{x - \mu}{\sigma} = \frac{5 - 25}{11} = -1.82.$$

- The Standard Normal Table shows that $P(z < -1.82) = 0.0344$.
- The probability that the student works for less than 5 hours per week is 0.0344.



So, 3.34% of college students with jobs worked for less than 5 hours per week. Because 3.34% is less than 5%, this is an unusual event.

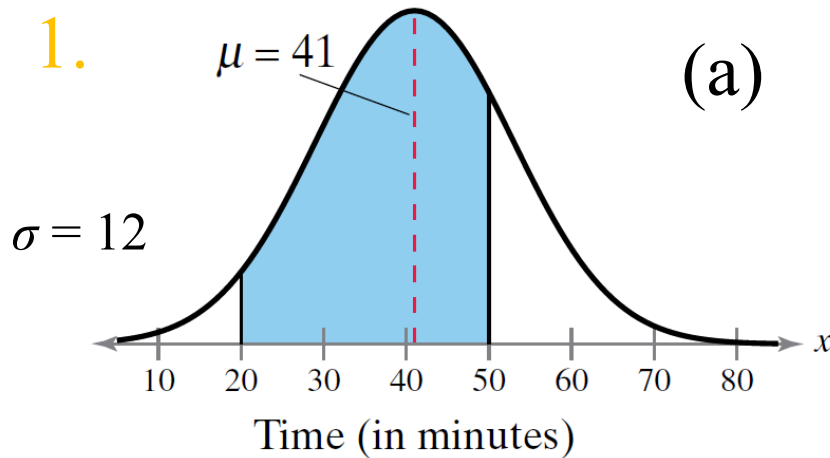
Example: Finding Probabilities for Normal Distributions

A survey indicates that for each trip to a supermarket, a shopper spends an average of 41 minutes with a standard deviation of 12 minutes in the store. The lengths of time spent in the store are normally distributed and are represented by the variable x . A shopper enters the store. (a) Find the probability that the shopper will be in the store for each interval of time listed below. (b) When 200 shoppers enter the store, how many shoppers would you expect to be in the store for each interval of time listed below? (*Adapted from Time Use Institute*)

1. Between 20 and 50 minutes
2. More than 35 minutes



Solution: Finding Probabilities for Normal Distributions



(a)

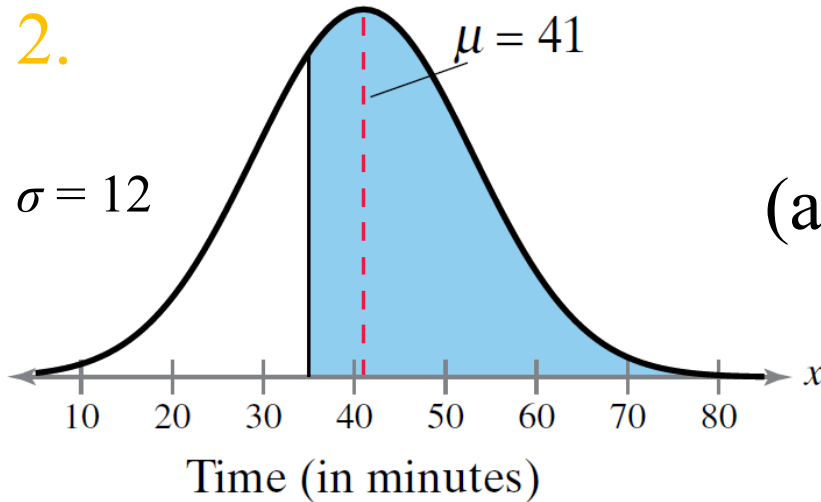
$$z_1 = \frac{x - \mu}{\sigma} = \frac{20 - 41}{12} = -1.75$$
$$z_2 = \frac{x - \mu}{\sigma} = \frac{50 - 41}{12} = 0.75$$

$$P(20 < x < 50) = P(-1.75 < z < 0.75)$$
$$= 0.7734 - 0.0401 = 0.7333$$

(b) When 200 shoppers enter the store, you would expect about $200(0.7333) = 146.66 \approx 147$ shoppers to be in the store between 20 and 50 minutes.

Solution: Finding Probabilities for Normal Distributions

2.



$$(a) \quad z = \frac{x - \mu}{\sigma} = \frac{35 - 41}{12} = -0.5$$

$$P(x > 35) = P(z > -0.5) = 1 - 0.3085 = 0.6915$$

(b) When 200 shoppers enter the store, you would expect about $200(0.6915) = 138.3 \approx 138$ shoppers to be in the store more than 35 minutes.

Example: Using Technology to find Normal Probabilities

In the United States, the numbers of physicians involved in patient care per state are normally distributed, with a mean of 280 physicians per 100,000 resident population and a standard deviation of 78 physicians per 100,000 resident population. You randomly select a state. What is the probability that the state has fewer than 300 physicians per 100,000 resident population? Use technology to find the probability. (*Adapted from National Center for Health Statistics*)

Solution: Using Technology to find Normal Probabilities

Minitab, Excel, StatCrunch, and the TI-84 Plus each have features that allow you to find normal probabilities without first converting to standard z-scores.

- must specify the mean and standard deviation of the population, as well as any x -values that determine the interval.
- $\mu = 280$
- $\sigma = 78$
- want $P(x < 300)$

Solution: Using Technology to find Normal Probabilities

TI-84 PLUS

```
normalcdf(-10000, 300,  
280, 78)  
.6011829115
```

MINITAB

Cumulative Distribution Function

Normal with mean = 280 and standard deviation = 78

x	P(X ≤ x)
300	0.601183

STATCRUNCH

Normal Calculator

Mean: 280 Std. Dev.: 78
 $P(x \leq 300) = 0.60118297$

EXCEL

	A
1	0.601182965

 ←=NORM.DIST(300,280,78,TRUE)

From the displays, you can see that $P(x < 300) \approx 0.601$.
The probability that the state has fewer than 300
physicians per 100,000 resident population is about 60.1%.